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Turning the page on energy poverty? Quasi-experimental evidence on education and energy poverty in Zimbabwe

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ABSTRACT

We investigate the long-term causal effects of education on energy poverty among households headed by men and women, leveraging the 1980 Zimbabwe school reform as a quasi-experiment. This reform, which significantly expanded educational opportunities for all children, serves as an ideal setting for assessing the impact of education on energy poverty within a regression discontinuity design, with age at reform determining exposure. Exposure to the reform resulted in a 2.08-year average increase in schooling and a 28.3 percentage point (pp) increase in the probability of completing secondary schooling. The reform's impact was more pronounced for women, increasing their schooling by 2.35 years (compared to 1.68 years for men) and for rural residents, with a 2.53-year increase (compared to 1.00 year for urban residents). A one-year increase in education reduces the risk of energy poverty by an estimated 8.56%, more pronounced among male-head families (5 pp) than among female-headed households (4.6 pp). This increased education enhances labour market outcomes, household economic status, infrastructure quality, and access to information and promotes progressive societal norms. These findings underscore education's crucial role in sustainable and gender-inclusive economic development, highlighting the potential of educational policy in reducing energy poverty and advancing gender equality in low-income countries.

1. Introduction

The interplay between education and energy poverty, characterised by inadequate access to essential modern energy services (International Energy Agency. IEA, International Renewable Energy Agency. IRENA, United Nations Statistics Division. UNSD, World Bank, and WHO, 2023), is crucial yet remains largely underexplored. Although several studies have noted that energy poverty negatively influences education (Banerjee et al., 2021), the inverse – particularly how education influences energy poverty – needs further exploration, especially in low-income countries. Most research on the nexus of education and energy poverty utilises macro-level data (Apergis et al., 2022). While informative, such data often obscure the mechanisms of influence and the varied benefits of education across different demographic and socioeconomic groups. The gap in understanding widens when considering gender disparities within household leadership, significantly impacting female-headed families. School reforms in low-income countries tend to disproportionately favour vulnerable groups (Glewwe and Kremer, 2006; Makate and Nyamuranga, 2024); however, it remains unclear whether these educational gains lead to similar reductions in energy poverty among households led by women compared to those led by men. This distinction is crucial as it addresses potential gender biases in the efficacy of educational policies and informs targeted interventions that can optimise the benefits of education across different household dynamics, ultimately enhancing policy effectiveness in reducing energy poverty.

This study is, therefore, vital in distinguishing the influence of the educational level of the household head – be it a woman or a man – on the household's energy poverty situation. We explore the detailed impacts of educational attainment on energy poverty among household heads, with particular focus on female-headed households, which face unique socioeconomic challenges and barriers to energy access in low-income countries (Ngarava et al., 2022). By emphasising these distinctions, we aim to highlight the critical role of education in empowering female-headed families and contributing to a more equitable reduction in energy poverty across different household dynamics. Our analysis also identifies specific areas where policy interventions can have a significant impact, guiding resource allocation to address gender disparities in education access and the effectiveness of policies to reduce energy poverty. This targeted approach underscores the need for

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tailored policy measures that bridge the educational gap and enhance energy access for female-headed households, ensuring a more inclusive and equitable pathway towards sustainable development.

In Zimbabwe, the situation of energy poverty mirrors a crisis that hampers access to energy, which, if not addressed, hinders socioeconomic development. The country grapples with an overall electricity penetration rate of 44%, with rural areas particularly affected, where access is only 20%, indicating glaring inequalities in energy provision (Climate Justice Central, 2022). The situation worsens as the total installed electricity capacity is merely 1200 MW, insufficient for a population of 14.9 million (Climate Justice Central, 2022). The quality and reliability of energy services are also major concerns, as power outages are common, and coal remains the predominant energy source. Such fragile circumstances undermine industrialisation and modernisation efforts, placing undue burdens on healthcare, education, and other essential services. At the same time, power supply shortages significantly affect the nation's economic performance, costing an estimated 6.1% of gross domestic product (GDP) annually (World Bank, 2023). Our analysis provides empirical evidence demonstrating that education is a crucial lever for achieving sustainable development goals (SDGs), particularly SDG 4 (Quality education) and SDG 7 (Affordable and clean energy) (United Nations, 2015, 2023). Improved educational outcomes can reduce energy poverty, aligning with SDG 4's emphasis on inclusive and equitable quality education as a foundation for improving lives and fostering economic development (United Nations, 2023). Improved education leads to better awareness and utilisation of sustainable energy resources, thus reducing energy poverty and advancing towards SDG 7's goal of universal access to affordable, reliable, and modern energy services. Moreover, our analysis explores the impact of education on sustainable energy access and considers gender dynamics within the household setting, providing empirical evidence on how education serves as a critical lever for sustainable development. This dual contribution underscores the integral role of education in advancing individual and societal well-being and catalysing progress towards sustainable development, particularly in bridging gender disparities and enhancing energy access. Understanding this relationship offers valuable insights for policymakers and stakeholders to leverage education as a strategic pathway to achieving the SDGs, especially in contexts marked by significant energy poverty and gender inequity.

In this paper, we investigate the long-term impact of Zimbabwe's 1980 education reform – a natural experiment that significantly expanded educational opportunities for all children in the country – on the energy poverty outcomes of individuals interviewed several years later. Using nationally representative data from the Zimbabwe Demographic and Health Survey (DHS), we explore the pathways through which education influences energy poverty. First, we consider education's effect on labour market outcomes, such as job stability and occupational choices, consistent with human capital theory (Becker, 2009), which posits that education enhances individual productivity and employability. Second, we consider education's influence on household economic status, looking at its correlation with improved asset wealth and infrastructure quality. Third, we assess education's impact on access to information, which is crucial for various economic decisions, including energy-related choices. This perspective aligns with the classic information economics theories, which assert that information asymmetry and acquisition costs broadly influence market behaviour (Stigler, 1961). Finally, we examine how education influences societal norms, including attitudes towards violence against women. This mechanism is rooted in the principles of behavioural economics, which explore how psychological, social, and cognitive factors influence economic decisions (Kahneman and Tversky, 1979). By fostering progressive societal norms and attitudes, education can indirectly mitigate energy poverty. This relationship underscores the interplay between education and broader societal values, illustrating how shifts in social norms can influence economic behaviours and choices. For instance, education can help reshape attitudes towards gender-based violence, leading to more equitable resource distribution and improved household energy use. This multifaceted role of education in promoting sustainable development highlights that societal values and norms are deeply intertwined with economic decisions, reinforcing the importance of holistic policy approaches in addressing energy poverty.

Our contributions to the economics literature on the intricate relationship between education and energy poverty are threefold. First, we use nationally representative micro-level data from the Zimbabwe DHS to examine the long-term effect of secondary school education on the risk of energy poverty. Second, we employ a regression discontinuity (RD) design, leveraging the exogenous variability in schooling prompted by the 1980 school reform, which expanded education opportunities for all children in Zimbabwe regardless of race or socioeconomic status. This school reform, which eliminated all bottlenecks that precluded indigenous children from enrolling or progressing to higher educational levels, serves as a quasi-experiment to evaluate the impact of education on energy poverty. Third, building on previous research indicating that women benefited more from the school reform than men, we conduct a heterogeneity analysis comparing the impact of education on energy poverty in female-headed versus male-headed households. This analysis distinctly differentiates between these household types, revealing the differential effects of education on energy poverty outcomes. Lastly, we detail the heterogeneous effects of education, and the underlying mechanisms through which increased secondary schooling influences energy poverty.

Our findings confirm that the school reform significantly increased educational attainment, particularly among women, individuals from low-asset wealth families and those living in rural communities. The reform contributed to an average increase of approximately 2.080 years in schooling and a 28.3 percentage point (pp) increase in the probability of secondary education. The impact was more significant for women, with an increase of 2.35 years in schooling compared to 1.68 for men. Rural residents experienced an average increase of 2.53 years in schooling compared to 1.00 years among urban residents. We find that increased schooling significantly reduces the prospect of energy poverty by an estimated 5.60 pp. (4.20 pp. among female-headed families and 5.60 pp. among male-headed families). Our analysis identified four mechanisms that potentially explain how education might reduce energy poverty: improved labour market outcomes, enhanced household economic status and infrastructure quality, improved access to information, and a reduction in the probability of holding harmful societal norms that condone violence against women.

The rest of the paper is structured as follows. Section 2 reviews the related literature; Section 3 provides a brief background of the 1980 school reform in Zimbabwe; Section 4 describes the data and empirical methods; Section 5 presents the results; Section 6 discusses the potential mechanisms, followed by a discussion of the results in Section 7. Section 8 provides a conclusion and policy considerations.

2. Literature review

The relationship between education and energy poverty has garnered increasing attention, revealing a range of impacts and underlying mechanisms that warrant further investigation. Despitethe evident link between energy poverty and education, especially in low-income countries, empirical evidence remains limited (Apergis et al., 2022). Apergis et al. (2022) suggest that increased education reduces energy poverty, although their reliance on macro-level data restricts a deeper understanding that micro-level analysis could provide. Acharya and Sadath (2019) further this narrative, indicating that education's impact on reducing energy poverty exceeds that of income, while also noting that gender and geographic location may influence this effect. Abbas et al. (2020) offer a detailed view, noting that while education generally correlates negatively with energy poverty in selected Asian countries, their country-specific analysis reveals mixed evidence. This observation highlights a critical gap in the literature, emphasising the need for

detailed studies that assess the long-term impact of education on energy poverty outcomes and detail the specific underlying mechanisms through which education influences energy poverty.

Gould et al. (2020) provide vital insights to understand the impact of education on energy-related behaviours, notably in adopting greener energy sources such as liquefied petroleum gas (LPG). Their comprehensive study, using data from nearly 10,000 households in Kerala and Rajasthan, India, uncovers a significant association between education and the adoption of LPG. Interestingly, while education strongly correlates with the uptake of this clean cooking fuel, it does not necessarily engender more favourable attitudes towards clean cooking fuels in general. This observation suggests a potential mechanism where education might influence energy poverty primarily through behavioural modifications rather than attitude shifts. However, a notable limitation of their study is its emphasis on associative relationships, which falls short of establishing a direct causal link between educational attainment and a comprehensive shift in attitudes towards sustainable energy practices. This gap underscores the need for further research to investigate these pathways, especially to understand the causal dynamics that underpin the long-term relationship between education and energy poverty in low-income countries.

Dagher et al. (2023) surveyed Lebanese households, finding that energy poverty inversely correlates with the education level of the household head. This observation means that higher educational attainment, often linked with higher income, reduces the incidence of energy poverty. This evidence from a crisis-stricken context underscores the significant role of education in alleviating energy poverty under severe economic conditions. Complementing this, Zhang et al. (2021) demonstrate the mediating role of academic performance in the relationship between energy poverty and children's well-being in China, showing that energy poverty adversely affects academic achievements and overall well-being. This finding underscores the long-term benefits of tackling energy poverty through educational improvements. Hassan et al. (2022) explore the environmental complexities of globalisation, education, and income inequalities, revealing that higher education levels correlate with reduced energy poverty by promoting environmental sustainability and reducing income inequality. Additionally, Crentsil et al. (2019) assess the determinants and drivers of multidimensional energy poverty in Ghana, identifying education as a crucial factor in reducing energy poverty through better energy-saving practices and increased adoption of clean energy technologies. Together, these studies highlight the multifaceted role of education in mitigating energy poverty across different contexts and mechanisms.

Recent research highlights the complex relationship where energy poverty adversely affects education. Banerjee et al. (2021) demonstrate this in their study of 50 developing nations from 1990 to 2017, revealing a correlation between reduced energy poverty and improved health and educational outcomes. Sule et al. (2022) focus on on the African context, illustrating how energy poverty adds to educational inequality, affecting educational access and quality directly, while others have linked it to stunted early development (Karmaker et al., 2022). Similarly, Bakehe (2022) investigates the broader effects of energy poverty on health in Cameroon showing that energy poverty increases the risk of respiratory illnesses and consequently impacts labour force participation, especially among women. Omar and Hasanujzaman (2021) discovered in Bangladesh that various dimensions of energy poverty have a detrimental impact on education, a view reflected in Rafi et al. (2021), who used data from India, where energy poverty hurts children's health and academic achievements. Research from Laos further confirms the negative consequences of energy poverty on schooling (Oum, 2019). These studies demonstrate how energy poverty can hamper educational success and quality, emphasising the need for holistic policies that address the subtle interplay between energy access, health, and education.

Our study significantly contributes to the literature by shifting the focus from examining the impact of energy poverty on education to exploring the long-term effects of education on energy poverty. This critical area, particularly in low-income countries, remains underexplored. Earlier studies have often found a negative correlation between education and energy poverty at the macro-level. However, they have not pinpointed the precise mechanisms, underscoring the need for micro-level data analysis. Building on this existing literature and addressing its gaps, our research shows how education, mainly secondary education, can offer sustainable solutions to energy poverty over the long term. Our exploration significantly advances the conversation about universal secondary education (USE) reforms in Africa and their broader implications, particularly among young women.

3. Transforming education: Zimbabwe's reform experience

3.1. Pre-reform educational landscape

Before Zimbabwe gained independence in 1980, educational disparities reflected the colonial legacy of racial and ethnic division. Under colonial rule, access to education was highly unequal, with significant differences in resources and opportunities available to different racial and ethnic groups. Schools for indigenous Zimbabweans were severely underfunded compared to those for white Zimbabweans, leading to substantial disparities in educational quality and outcomes. The gross enrolment rate in primary education was below 65% in the 1970s, with less than half of those students transitioning to secondary school (Dorsey, 1989; Kanyongo, 2005). This meant that a significant proportion of the population, particularly those in rural areas and from disadvantaged backgrounds, had limited access to even basic education. These disparities underscored the urgent need for comprehensive educational reforms. The inequitable distribution of distribution of educational resources and opportunities not only perpetuated social and economic inequalities but also hindered the nation's overall development potential. This pre-independence educational landscape created a pressing imperative for the new Zimbabwean government to prioritise educational reform as a key component of its nation-building efforts.

3.2. The education act of 1980 and reforms

After independence, the government prioritised education as a crucial part of nation-building. The Education Act of 1980, a significant piece of legislation, reformed the education system in Zimbabwe and is widely regarded as one of the most successful school reforms in sub-Saharan Africa (SSA) (Dorsey, 1989; Kanyongo, 2005; Nhundu, 1992). The act set the entry age into formal primary education at six years, with a seven-year curriculum culminating in Grade 7 exams. After primary school, students transitioned to secondary education, which officially began at age 13. This phase included a four-year Ordinary Level followed by a two-year Advanced level. The reforms aimed to standardise educational opportunities for all children, regardless of race or socio-economic status (Edwards and Tisdell, 1990).

To support these reforms, the government abolished tuition fees and invested heavily in educational infrastructure, significantly increasing the transition rate from Grade 7 to Form 1. The transition rate increased to approximately 85% by the late 1980s from sub-60% in the late 1970s (Maravanyika, 1990). Furthermore, the reforms enabled older individuals who had previously dropped out of the education system to resume their studies. In Zimbabwe, children typically start primary school at six, advancing through a seven-year program before transitioning to secondary school. The policy reforms significantly impacted children aged 13 in 1980, as they were within the secondary school starting age and thus in the best position to benefit from the changes. The relaxation of the overage policy also allowed 14–15-year-olds to benefit from the reforms. However, previous studies analysing the impact of the school reform in Zimbabwe using DHS data showed that those aged 16 and older were less likely to be affected by the reforms (Agüero and Bharadwaj, 2014; Fenske, 2015; Makate and Makate, 2018;

Makate and Nyamuranga, 2023).

3.3. Post-reform developments and impact

The comprehensive educational reforms of 1980 marked a transformative era for Zimbabwe's education system. Post-reform, the gross enrolment rate in primary education soared to over 90% by 1990, with the transition rate to secondary education also experiencing a significant increase to approximately 70% (Edwards and Tisdell, 1990). Secondary school enrolment saw the most dramatic growth, from 66,215 in 1980 to over 704,000 by 1990, a staggering rise of approximately 964% (Makate and Nyamuranga, 2023). This substantial increase can be attributed to the introduction of free primary education, the abolition of tuition fees, and significant government investment in educational infrastructure and teacher training. These measures collectively created a more inclusive and accessible education system, particularly benefiting those in rural areas and from disadvantaged backgrounds.

The reforms not only increased enrolment rates but also led to substantial improvements in educational attainment and quality. Government investment in teacher training and infrastructure, coupled with curriculum development, significantly enhanced literacy rates and educational outcomes. By 1990, the literacy rate for individuals aged 15 and above had increased from approximately 62% in 1982 to over 80%, reflecting the success of these reforms in enhancing educational outcomes across the country (UNESCO Institute for Statistics, 2024; World Bank, 2024). These educational advancements translated into significant economic benefits, as a better-educated workforce contributed to economic growth and development. The enhanced educational opportunities enabled more individuals to pursue higher education and vocational training, thereby strengthening the nation's skilled labour pool (Edwards and Tisdell, 1990). Additionally, the reforms played a crucial role in reducing gender disparities in education, with female enrolment rates in primary and secondary education approaching parity with male enrolment rates by the late 1980s (Kanyongo, 2005). The 1980 education reforms had a lasting impact that extended beyond immediate enrolment and transition rates. The improved educational landscape fostered a generation of individuals better equipped to participate in and contribute to the socio-economic development of Zimbabwe. This transformation underscored the importance of sustained investment in education as a cornerstone of national development (World Bank, 2020).

4. Data and empirical strategy

4.1. Data and measures

We use data from the Zimbabwe DHS collected in 1999, 2005/2006, 2010/2011, and 2015. These surveys, aimed at providing updated estimates of basic demographic and health indicators, are based on the nearest preceding national census to ensure representative sampling (Zimbabwe National Statistics Agency, Z., and ICF International, 2016). The DHS, conducted in over 90 countries, is renowned for its comprehensive demographic and health data on women, children, and men (The DHS Program, 2023). It includes household-level information such as asset ownership, housing conditions, and access to utilities, which are crucial for assessing socioeconomic status and energy poverty indicators. For women aged 15–49, the survey covers a range of topics including demographic background, birth history, family planning, maternal health, childbirth, marriage, sexual activity, employment, educational attainment, malaria, HIV/AIDS, and sexually transmitted infections (STIs), and domestic violence. The man's questionnaire, administered to men aged 15–54, is similar but excludes detailed reproductive, maternal, and child health questions.[1]

Table 1 presents sample-weighted summary statistics for selected variables of men and women, stratified by reform exposure status for those aged 5–24 in 1980, which is our preferred age bandwidth. The average years of completed schooling for individuals in our preferred bandwidth of 5–24 years in 1980 was 8.37 years. There was an approximate 2.53-year difference in years of completed schooling between those exposed to the school reform (9.07 years) and those not exposed (6.54 years). The average secondary schooling completion rate was higher among the exposed (70%) compared to the non-exposed group (34%). The proportion of females was slightly higher among the exposed group (0.58) than the non-exposed group (0.54). The average age in 1980 was 11.99 years, with the exposed group having an average age of 9.18 years and the non-exposed group 19.30 years. At the time of the survey, the average age was 38.85 years, with the exposed group averaging 36.46 years and the non-exposed group averaging 45.07 years. The age of the head of household averaged 43.18 years, with the exposed group averaging 41.42 years and non-exposed group 47.76 years. Marital status showed slight differences, with 5% of the exposed group never married compared to 1% of the non-exposed group. Household size, female head of household status, and religious affiliation showed minor differences between the groups. For occupation, 12% of the overall sample worked in skilled manual work, with 11% in the exposed group and 13% in the non-exposed group. Land ownership was slightly higher among the non-exposed group (0.74) than the exposed group (0.68). The frequency of listening to the radio was higher among the exposed group (0.64) than the non-exposed group (0.60). The proportion of individuals with low asset wealth (poorer or poorest) was marginally higher among the non-exposed group (0.37) than the exposed group (0.33). The rural resident proportion was higher among the non-exposed group (0.67) than the exposed group (0.62). The distribution across regions showed minor variations between the two groups.

4.2. Measuring energy poverty

The economic literature on energy poverty measurement has evolved significantly, reflecting diverse approaches across different income contexts. A recent comprehensive review of the literature highlights the evolving definitions and measurements of energy poverty, emphasising the multidimensional nature of the concept (Lu and Ren, 2023). The United Nations' modern definition characterises energy poverty as the inability to acquire a sufficient amount of reliable and high-quality energy that is affordable, environmentally friendly, and safe (United Nations, 2023).

In high-income countries, energy poverty measurements combine objective criteria, such as affordability and energy efficiency, with subjective assessments, including household perceptions of energy access (Awaworyi Churchill and Smyth, 2021; Churchill et al., 2020; Hailemariam et al., 2021). A systematic review by Siksnelyte-Butkiene et al. (2021) identified 71 composite indicators, illustrating the complexity of energy poverty. Nonetheless, different regions' diverse

[1] A detailed description of the survey methodology for the Zimbabwe DHS is accessible here (Zimbabwe National Statistics Agency and ICF International, 2016). The report provides comprehensive information on the sampling design, data collection procedures, and data analysis methods used in the survey. This detailed methodology ensures the reliability and validity of the survey findings, making it a valuable resource for understanding demographic and health trends in Zimbabwe.

Table 1
Weighted summary statistics of individuals (men and women) by treatment status in Zimbabwe.

	Overall sample		Exposed to school reform		Not exposed to school reform	
Variables	Obs.	Mean	Obs.	Mean	Obs.	Mean
Female	16,396	0.57	11,447	0.58	4949	0.54
Age in 1980	16,396	11.99	11,447	9.18	4949	19.30
Age at survey date	16,396	38.85	11,447	36.46	4949	45.07
Age of head of household	16,393	43.18	11,445	41.42	4948	47.76
Years of schooling	16,396	8.37	11,447	9.07	4949	6.54
Completed secondary school	16,396	0.60	11,447	0.70	4949	0.34
Marital status						
Never married	16,396	0.04	11,447	0.05	4949	0.01
Widowed	16,396	0.10	11,447	0.08	4949	0.13
Divorced	16,396	0.04	11,447	0.05	4949	0.04
Separated	16,396	0.03	11,447	0.03	4949	0.03
Household size (=1 if 2 members)	16,396	0.08	11,447	0.07	4949	0.08
Female head of household	16,396	0.33	11,447	0.33	4949	0.32
Religion affiliation						
Apostolic member	16,394	0.25	11,446	0.26	4948	0.21
Protestant	16,394	0.16	11,446	0.16	4948	0.17
Occupation						
Skilled manual work	16,396	0.12	11,447	0.11	4949	0.13
Unskilled manual work	16,396	0.06	11,447	0.06	4949	0.05
Domestic work	16,396	0.03	11,447	0.03	4949	0.03
Services work	16,396	0.06	11,447	0.06	4949	0.06
Owns land	3118	0.69	2805	0.68	313	0.74
Frequently listen to radio	16,391	0.63	11,438	0.64	4947	0.60
Low asset wealth (poorer and poorest)	16,396	0.34	11,447	0.33	4949	0.37
Rural resident	16,396	0.64	11,447	0.62	4949	0.67
Region/province						
Manicaland	16,396	0.12	11,447	0.12	4949	0.13
Mashonaland central	16,396	0.11	11,447	0.11	4949	0.11
Mashonaland east	16,396	0.10	11,447	0.10	4949	0.10
Mashonaland west	16,396	0.12	11,447	0.13	4949	0.11
Matabeleland north	16,396	0.06	11,447	0.06	4949	0.06
Matabeleland south	16,396	0.05	11,447	0.05	4949	0.05
Midlands	16,396	0.13	11,447	0.13	4949	0.13
Masvingo	16,396	0.11	11,447	0.11	4949	0.13
Harare	16,396	0.15	11,447	0.16	4949	0.12
Bulawayo	16,396	0.06	11,447	0.05	4949	0.06

Notes: The table presents the survey-weighted estimates and number of observations for men and women who were aged 5–24 years in 1980 and stratified by whether they were exposed (aged 13 years and younger) to school reform or not (aged 14 years and older). The last column reports the p-value, calculated from a standard *t*-test, between the means for individuals exposed to the school reform and those who were not. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys.

socioeconomic and energy contexts challenge these measurements.

In low-income countries, energy poverty measures focus on energy access and quality. Studies show that energy poverty in these regions often intertwines with broader issues of economic development, sustainable finance, and resource management (Raghutla and Chittedi, 2022). In our analysis, we employ the multidimensional energy poverty index (MEPI), initially proposed by Nussbaumer et al. (2012). This index is particularly suitable for low-income countries, given the typical data constraints in such settings. With the widespread use of the DHS, recent studies in low-income countries have relied on this micro-level data to measure energy poverty using the MEPI (Abbas et al., 2022; Munyanyi and Churchill, 2022; Nussbaumer et al., 2012).

The DHS asks respondents several questions regarding: (i) ownership of a radio or television, (ii) types of fuel used for everyday cooking, (iii) access to electricity, (iv) ownership of a mobile phone or household telephone, and (v) ownership of electrical appliances such as a refrigerator. We classify households as energy-poor on each indicator if they indicate having no ownership or access. A summary of the indicators alongside their weights is provided in Table A2 of the supplementary appendix. The weights' choice and relative importance in the context of low-income countries, including Zimbabwe, have been discussed elsewhere (Koomson and Danquah, 2021; Munyanyi and Churchill, 2022; Nussbaumer et al., 2012).

We follow previous studies in low-income countries and use the MEPI as our primary measure of energy poverty (Abbas et al., 2022; Munyanyi and Churchill, 2022; Nussbaumer et al., 2012). The MEPI assigns weights to five energy-use forms: cooking, lighting, communication, entertainment/education, and household appliances. The weighting system reflects the diverse ways energy is accessed and used. The MEPI also considers the affordability of energy, which is measured through survey questions that assess households' ability to use and afford energy-requiring appliances. This approach aligns with the broader discussions on energy rationing in the literature, which frame energy poverty as a lack of access to essential energy-requiring appliances and the inability to afford them. Consequently, families that cannot access or afford essential appliances, including those for communication and entertainment, are considered energy-poor (Nussbaumer et al., 2012). The algebraic representation of the energy deprivation score for each household can be formalised using the details on the k indicators (Nussbaumer et al., 2012).

$$D_i = \sum_{j=1}^{k} w_j I_{ji} \tag{1}$$

where, D_j is the deprivation score for household i, w_j is the weight attached to indicator j, such that $\sum_{j=1}^{k} w_j = 1$, and I_{ji} is an indicator variable that equals 1 if household i is deprived in the indicator j and zero otherwise. This deprivation score is then used to define the multidimensional energy poverty status of the household based on a predetermined cut threshold of c with a household being regarded as energy poor if $D_j \geq c$. In this paper, we follow Nussbaumer et al. (2012) and classify a family to be multidimensionally energy deprived or poor if its deprivation score is greater than or equal to 0.50 i.e., $D_i \geq 0.50$. In a sensitivity check, we consider the instance when the threshold is set to 0.3 and 0.7 instead. The threshold of 0.3 has been supported in the literature and is thought to adequately capture the acutely energy-deprived families (Alkire and Santos, 2014; Nussbaumer et al., 2012). We also use the multidimensional energy deprivation score ($MEPI\ score_j$) in its continuous form.

In Zimbabwe, as in many other low-income countries, cooking accounts for a considerable amount of household energy consumption and, therefore, is assigned a greater weight of 0.40 in the MEPI calculation to reflect greater deprivation (International Energy Agency, 2019; Munyanyi and Churchill, 2022; Nussbaumer et al., 2012). Households are energy-deprived if they use unimproved cooking fuels like firewood, charcoal, animal dung and agricultural waste (Nussbaumer et al., 2013). In Zimbabwe, due to the intermittent availability of electricity, even relatively wealthy families and urban residents have increased their reliance on unimproved energy sources for cooking, such as firewood and charcoal (UNDP, 2020).

Aside from cooking, access to electricity for lighting is essential, so the second highest weight of 0.20 is assigned. While the Zimbabwean government has made remarkable efforts to improve access to electricity in the country, inequalities in access persist. According to the World Bank, 53% of Zimbabwe's population had access to electricity in 2020 and 49% in 2021, with 86% living in urban areas and 37% in rural areas) (World Bank, 2022). Even in metropolitan areas and among the

relatively affluent, intermittent availability of electricity has meant substituting cleaner energy sources like electricity for less clean energy sources like firewood and charcoal (UNDP, 2020). The remaining fraction is equally apportioned to indicators reflecting the usage of electrical appliances and telecommunications (Nussbaumer et al., 2012). Table A3 shows the evolution of energy poverty over the years. Between 1999 and 2015, there was a notable decrease in energy poverty, as indicated by changes in the MEPI status. In 1999, 69.65% of the population was classified as multidimensionally energy-deprived, reduced to 60.07% by 2015. This decline in energy poverty occurred alongside improvements in educational attainment. The proportion of the population that completed secondary school increased from 51.37% in 1999 to 75.06% in 2015, while the average years of completed schooling rose from 7.71 to 9.65 years (see Table A3).

4.3. Measuring educational attainment

Our primary independent variable is educational attainment, measured as the years of completed schooling, and a dummy variable representing secondary school completion as recorded at the time of the survey. This dummy variable, assigned a value of one for respondents who have completed secondary school and zero otherwise, highlights explicitly the educational advancement occurring at the margin between primary and secondary schooling.

4.4. Empirical approach

4.4.1. Identification strategy

To consistently estimate the causal impact of education on energy poverty, we employ an RD design, leveraging the exogenous variation in schooling prompted by Zimbabwe's 1980 educational reform. This reform mandated increased access to secondary education for individuals aged 13 or younger at the time, creating an explicit cut-off for exposure that aligns well with the RD design methodology. Our identifying assumption is that individuals just above and below this age threshold are comparable in terms of observable characteristics, differing primarily in their exposure to the reform. This approach mimics a randomised treatment assignment, enhancing the validity of our causal inference. The age-specific component of the reform provides the source of exogenous variation in schooling to reliably test for the causal effect of education on energy poverty outcomes. This identification strategy has been successfully applied in previous studies analysing the impact of the Zimbabwe school reform on other outcomes (Agüero and Bharadwaj, 2014; Makate and Makate, 2018; Makate and Nyamuranga, 2023, 2024).

To examine the impact of education on energy poverty, we start with a very basic formulation expressed as follows:

$$\mathrm{EnPov_i} = a_0 + a_1 \mathrm{EducYears_i} + \delta X_i + \varepsilon_i \tag{2}$$

where, $\mathrm{EnPov_i}$ is the outcome variable capturing energy poverty, and $\mathrm{EducYears_i}$ denotes the years of education completed by the ith individual as observed by each survey date. The term X_i is a vector of control variables, which includes survey month fixed effects, survey year fixed effects, urban residence indicator, region fixed effects, and region-birth-cohort fixed effects. Survey month fixed effects control for seasonal variations in energy consumption and expenditure, which can influence the relationship between education and energy poverty. Seasonal fluctuations in household energy consumptions are well documented in the literature (Paxson, 1993; Stephens Jr and Unayama, 2011). These variations can introduce bias if not accounted for, affecting the estimation of the relationship between educational attainment and subsequent energy poverty status. Additionally, household income levels in developing countries are often subject to temporal shocks due to seasonal cycles in economic activities like agriculture (Beegle et al., 2011). These income fluctuations can impact both educational attainment and long-term energy poverty outcomes. Therefore, including survey month fixed effects helps mitigate the potential confounding influence of such time-varying income shocks. Survey-year fixed effects account for year-specific shocks and trends, such as economic cycles and policy changes, that may influence education and energy poverty (Angrist and Pischke, 2009). The urban residence indicator addresses structural differences between urban and rural areas regarding educational resources and energy infrastructure, which are critical in understanding disparities in outcomes (Beegle et al., 2011). Region-fixed effects control for unobserved heterogeneity across geographic areas, which can significantly influence both educational attainment and economic outcomes. Region-birth-cohort fixed effects capture region-specific trends over different birth cohorts, isolating the impact of the educational reform from other regional factors. Furthermore, measuring energy poverty outcomes through household surveys can be susceptible to systematic measurement error or reporting biases that vary by the survey's timing (Bound et al., 2001). Survey month fixed effects can help absorb such potential measurement errors, enhancing the precision and robustness of the estimates.

We can estimate eq. (2) using ordinary least squares (OLS) to determine the coefficient of interest, a_1, measuring the association between education and energy poverty. Endogeneity presents a critical challenge in consistently estimating the coefficient a_1, which represents the effect of educational attainment on energy poverty. Several factors contribute to this endogeneity. First, omitted variable bias is likely to occur if there are unobserved variables that simultaneously affect both educational attainment and energy poverty but are not included in the control vector X_i. Second, measurement error in variables like $\mathrm{EducYears_i}$ or $\mathrm{EnPov_i}$ could lead to a biased estimation of a_1. Third, the issue of simultaneity could further complicate the endogeneity problem, particularly if energy poverty has a reciprocal effect on educational attainment. These factors necessitate an econometric approach robust to these issues for a reliable estimation of a_1.

To address endogeneity in estimating the impact of education represented by the coefficient a_1, we leverage the exogenous variation in schooling prompted by Zimbabwe's 1980 school reform, which significantly expanded secondary education access (Agüero and Bharadwaj, 2014; Makate and Nyamuranga, 2023). Given that children in Zimbabwe typically start secondary school at age 13, this age is the natural cut-off for reform exposure. Accordingly, we classify individuals 13 years or younger in 1980 as exposed to the reform, while those older are non-exposed. This age-based criterion forms the core of our RD design, allowing us to consistently evaluate the causal effect of schooling on energy poverty outcomes several years later in the life course. The crux of our identification strategy is the plausible assumption that individuals on either side of the age 13 cut-off are observationally equivalent, except in their exposure to educational reform. If this assumption holds, our RD design serves as a quasi-random treatment assignment, thus bolstering the validity of our causal inference.

Within this RD framework, and as in previous literature, we provide both reduced-form (RF) estimates (i.e., sharp RD) and instrumental variable (IV) estimates (i.e., fuzzy RD) for all outcomes of interest. Our specification follows a basic RD formulation:

$$\mathrm{EnPov_i} = \alpha + \beta \mathrm{EducReform_i} + g(a80_i) + \gamma X_i + u_i,$$
$$\forall a80_i \in (13-h, 13+h), \tag{3}$$

where, $\mathrm{EnPov_i}$ is the dependent variable measuring energy poverty assessed several years later after the school reform from 1999 to 2015, $\mathrm{EducReform_i}$ is a dummy indicator for exposure to the 1980 school reform, which equals one if the respondent was aged 13 years or younger in 1980 and zero otherwise, $a80_i$ is the running variable – the respondent's age at the time of school reform and calculated as the difference between month-year of reform and month-year of birth, and h is the bandwidth around the age 13 cut-off point. In the econometric

framework, we introduce $g(a80_i)$ as a segmented linear control function to condition for age-specific idiosyncratic trends around the age-13 discontinuity, both pre- and post-policy implementation. Consistent with extant RD literature (Imbens and Lemieux, 2008), our empirical approach leverages local linear regression estimators on either side of the discontinuity. The empirically chosen optimal bandwidth spans a nine-year interval centred around the age 13 cut-off. We also consider starting with a broader bandwidth of 10 and gradually decreasing in one-year decrements to smaller bandwidths of 5 years on either side, consistent with the RD literature (Imbens and Lemieux, 2008). We follow Makate and Nyamuranga (2023) and cluster standard errors at the region and age in 1980 levels to account for possible different intensities of school construction across the country's ten regions or provinces. Thus, clustering captures or accounts for age cohort-specific spatial correlations across regions.

4.4.2. Preliminary and robustness checks

Following the RD literature, we assess the key identifying assumption that the outcome variable is smooth around the cut-off (Lee and Lemieux, 2010). The validity of the RD analysis rests on the belief that the continuity assumption, which states that potential outcomes are continuous around the discontinuity threshold value, holds (Hahn et al., 2001). However, as the continuity assumption is based on unobservable outcomes, it is technically untestable (Lee and Lemieux, 2010). Following the recommendations in the RD literature, we provide indirect assessments of this assumption. First, we check for the possibility of manipulating the running variable and if it exhibits any discontinuities around the cut-off point (Lee and Lemieux, 2010). To this end, we present a histogram of the assignment variable and conduct the McCrary test (McCrary, 2008).[2] The results for the McCrary test are presented in Fig. 1 and do not reveal any evidence of manipulation in the assignment variable. The density distribution of the respondents' age in 1980 is continuous around the age 13 threshold. The latter observation together with the nature of the assignment variable, which is less likely to be manipulated, we are confident that assignment into the treatment cohort is purely random. The histogram of the assignment variable presented in Fig. A1 shows a smooth distribution of the running variable around the age 13 threshold.

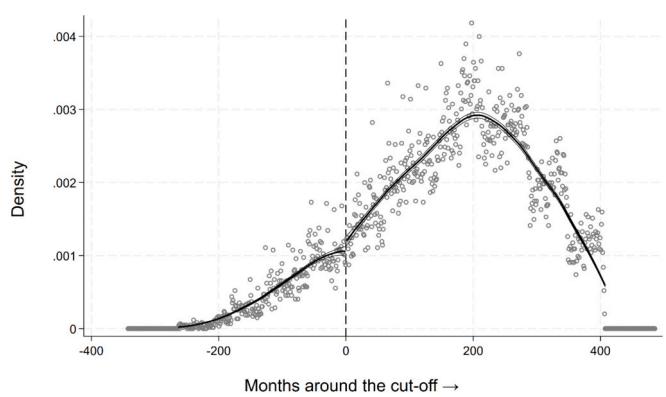

Fig. 1. McCrary Density Test. Notes: Data comes from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health survey. This figure plots the results of the McCrary density test of whether there exists a discontinuity in the density distribution of the running variable, months around the age 13 cut-off (i.e., zero represents born in 1967). The fitting procedure allows for discontinuity at zero, with a bin size of one. Bandwidth standard errors, and the density plot were produced using McCrary's DCdensity package, following McCrary (2008).

Second, we assess the absence of a jump in the other covariates we included in the regression analysis within the neighbourhood of the running variable. We present the results for this analysis as supplementary materials in Figs. A2-A4. Figs. A2-A4 present the local means of each variable in one-month bins plotted against the running variable. The results displayed in these graphs show no apparent discontinuities in the predetermined variables we used. Additionally, we provide in Table A1 the magnitudes of the jumps in the predetermined covariates alongside their *p*-values. The coefficients reported in Table A1 are generated using regression equations of the type in eq. (3), with each covariate as the dependent variable and the reform exposure dummy indicator as the key independent variable. In these regressions, we also include indicators for the linear age slopes below and above the cut-off point as additional controls.

Third, we conducted placebo experiments by simulating the school reform as if it had been enacted in 1974, 1975, 1976, or 1977, four to six years earlier than the actual enactment in 1980. The concern is that exposure to school reform may be picking up on other structural changes correlated with energy poverty apart from the 1980 school reform. Thus, the discontinuity we observe at age 13 may not be unique, rendering our conclusions invalid. The results of these checks are presented in Tables A4 and A5 and reported for all the outcome variables we used in the study. The placebo reforms' coefficients lacked statistical significance, further supporting our identification strategy and the robustness of our primary estimates.

5. Results

5.1. School reform and education outcomes

We first examine the impact of school reform on the educational outcomes of men and women. The Zimbabwe DHS reports information on the month and year of birth data for both men and women. Using this data, we present the RD treatment effects of the school reform on years of completed schooling and secondary school completion in month-of-birth bins around the cut-off point (January 1967 or age 13 years and younger) for men and women. The top panel of Fig. 2 shows a distinct jump in educational attainment for women, evident in both the years of completed schooling and the secondary school completion rate. The bottom panel, showing the graphs for men, exhibits a less pronounced discontinuity in educational outcomes. These contrasting results suggest that the reform substantially impacted women more than men. This discrepancy may be attributed to pre-existing patriarchal norms favouring boys' education over girls prior to independence (Chikunda and Chikunda, 2016; Gordon, 1994), leading to comparatively better

[2] The McCrary test is a diagnostic tool used in RD designs to test for manipulating the running or assignment variable around the cutoff point. Specifically, it examines whether the density of the running variable is continuous at the threshold, which is crucial for the validity of the RD design. If there is a discontinuity in the density at the cutoff, it suggests that individuals may have manipulated their position relative to the cutoff, thereby compromising the quasi-random assignment of treatment. This test, developed by Justin McCrary (2008), is implemented by estimating the density of the running variable on either side of the threshold and testing for a statistically significant difference.

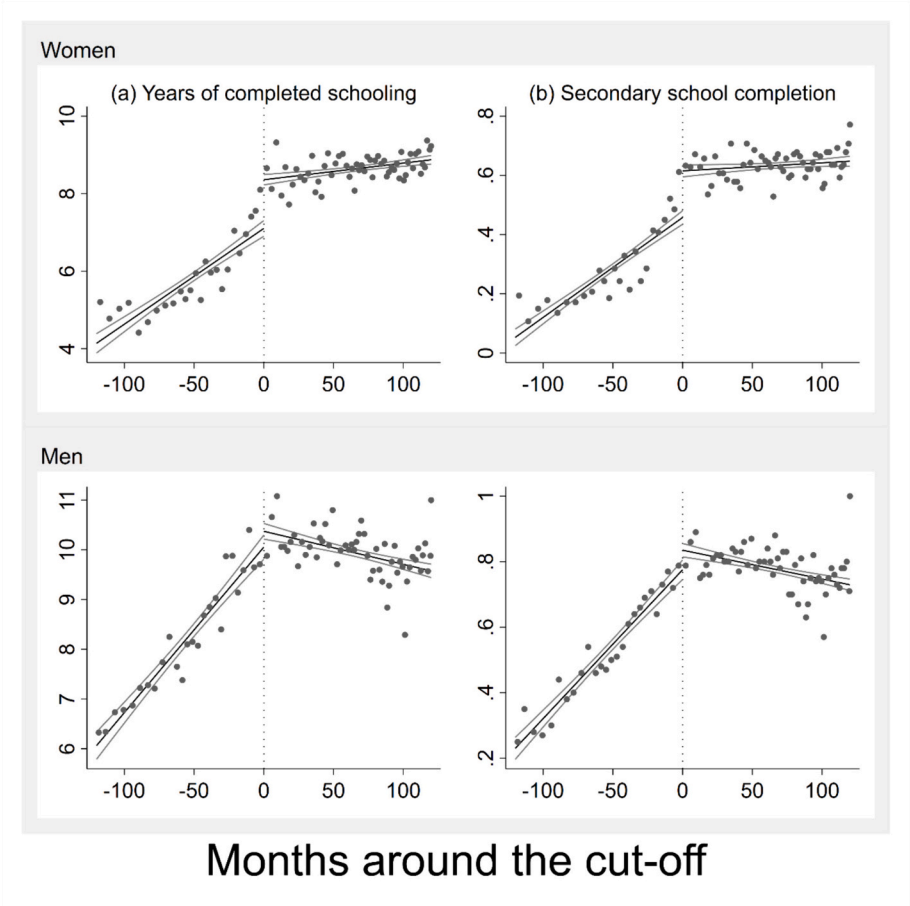

Fig. 2. The RD treatment effect of the 1980 school reform on years of completed schooling and secondary school completion among women and men in Zimbabwe. The panels provide plots of the average years of schooling in monthly bins against the month-year of birth of being born in 1967. The point of discontinuity is centred on individuals born in 1967 – those who were 13 years old at the time of the school reform in 1980. The x-axis measures the number of months relative to this pivotal year when the respondent was 13 years old. Gray lines represent the 95% confidence intervals. The complete definitions of the variables of interest are provided in appendix. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys.

initial access to education for males.

Visualising the reform's effect on educational outcomes reveals a favourable RD treatment effect. However, regression analysis allows for a more detailed understanding. Table 2 presents the findings from the first stage regression analysis of the policy impact on educational outcomes, which employs eq. (3) with years of schooling or secondary school completion as the dependent variable. This first stage regression model includes additional controls for linear polynomials of the distance to the age threshold on either side from above and below the cut-off point (i.e., pre- and post-policy linear age trends), an indicator for being female, survey month fixed effects, year of survey fixed effects, and region fixed effects. Furthermore, we use adjusted survey probability weights to address any potential bias from pooling data from multiple rounds of the Zimbabwe DHS. To validate the consistency of our results, we report over a range of bandwidths from 10 to 5 years on each side of the cut-off point. We also examine whether the reform had heterogenous effects based on gender, and place of residence (rural or urban).

Table 2 presents the RD treatment effects on the years of completed schooling and the probability of completing secondary school for all men and women interviewed in 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe DHS. The top panel (panel a) displays estimates of the RD treatment effects on years of completed schooling for both genders across several bandwidths, ranging from 10 to 5 years on either side of the cut-off point. Overall, the reform increases years of schooling by approximately 1.67 years in the narrower bandwidth and about 2.05 years in the 10-year bandwidth, with all estimates being statistically significant at the 1% level. A more substantial effect appears within a 9-year bandwidth, where the reform associates with a 2.08-year overall increase in years of schooling. Notably, the reform's impact is more pronounced in the rural subsample, increasing years of schooling by an estimated 2.53 years compared to 1.00 years in the urban subsample within the 9-year age bandwidth. The impact on female education is also more significant than on men, with females acquiring an average of 2.35 years compared to 1.68 years for males in the 9-year age bandwidth, all statistically significant at the 1% level. Similarly, the reform's impact on secondary school completion ranges from 0.23 pp in narrower bandwidths to 0.27 pp. in broader bandwidths. The rural (0.33 pp) and female (0.31 pp) subsamples exhibit more pronounced policy effects in the 9-year bandwidth, each statistically significant at the 1% level.

5.2. Education and energy poverty

In this section, we investigate the long-term effect of school reform on energy poverty outcomes several years after the reform in the lives of individuals in Zimbabwe, focusing on female- and male-headed households. Our identification strategy hinges on the idea that an individual's age at the time of the 1980 school reform determines their exposure to it. Specifically, those within the official secondary school starting age of 13 years and younger were more likely to have attended secondary school than their counterparts aged 16 and older, who were outside the typical secondary school starting age in Zimbabwe. Initially, we present

Table 2
Regression discontinuity treatment effects of school reform on educational outcomes of men and women in Zimbabwe.

	Number of years on either side of the cut-off point.											
	10		9		8		7		6		5	
Dep. variable: Years of completed schooling												
a) *Overall sample*												
Exposure to school reform	2.052***	(0.272)	2.082***	(0.265)	2.055***	(0.227)	1.962***	(0.177)	1.721***	(0.169)	1.674***	(0.250)
Observations	9720		8749		7676		6629		5631		4550	
b) *Urban subsample*												
Exposure to school reform	1.032**	(0.343)	0.999**	(0.327)	1.097***	(0.228)	1.032***	(0.206)	0.928***	(0.262)	0.655	(0.411)
Observations	3335		2984		2627		2252		1913		1589	
c) *Rural subsample*												
Exposure to school reform	2.435***	(0.289)	2.532***	(0.286)	2.419***	(0.299)	2.327***	(0.238)	2.013***	(0.212)	2.191***	(0.320)
Observations	6385		5765		5049		4377		3718		2961	
d) *Female subsample*												
Exposure to school reform	2.344***	(0.249)	2.348***	(0.251)	2.274***	(0.231)	2.299***	(0.183)	1.954***	(0.064)	1.898***	(0.189)
Observations	6268		5632		4920		4273		3613		2905	
e) *Male subsample*												
Exposure to school reform	1.620***	(0.196)	1.679***	(0.179)	1.653***	(0.122)	1.396***	(0.060)	1.322***	(0.189)	1.375***	(0.188)
Observations	3452		3117		2756		2356		2018		1645	
Dep. variable: Completed secondary school												
a) *Overall sample*												
Exposure to school reform	0.274***	(0.028)	0.283***	(0.026)	0.273***	(0.017)	0.271***	(0.017)	0.250***	(0.020)	0.230***	(0.035)
Observations	9720		8749		7676		6629		5631		4550	
b) *Urban subsample*												
Exposure to school reform	0.164**	(0.051)	0.170**	(0.055)	0.176***	(0.049)	0.192***	(0.048)	0.199***	(0.048)	0.113	(0.073)
Observations	3335		2984		2627		2252		1913		1589	
c) *Rural subsample*												
Exposure to school reform	0.321***	(0.019)	0.334***	(0.012)	0.314***	(0.003)	0.304***	(0.002)	0.268***	(0.007)	0.299***	(0.023)
Observations	6385		5765		5049		4377		3718		2961	
d) *Female subsample*												
Exposure to school reform	0.309***	(0.012)	0.310***	(0.007)	0.297***	(0.001)	0.306***	(0.008)	0.282***	(0.014)	0.262***	(0.012)
Observations	6268		5632		4920		4273		3613		2905	
e) *Male subsample*												
Exposure to school reform	0.224***	(0.035)	0.241***	(0.038)	0.226***	(0.025)	0.205***	(0.038)	0.185***	(0.053)	0.170***	(0.046)
Observations	3452		3117		2756		2356		2018		1645	

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The estimates in the rural and urban columns are derived from analyses conducted with the defined sample of men and women who were aged 5–24 years in 1980. Standard errors for coefficient estimates are displayed in parentheses and are clustered at the birth month-year cohort level. We excluded individuals who were partially exposed to the school reform, especially those aged 14–15 in 1980, in line with previous studies that employed the same identification strategy. All regressions presented in panels a-e include the following variables as additional controls: pre-policy and post-policy trends, survey fixed effects and region fixed effects. All regressions are weighted using sample weights that have been adjusted to account for pooling across multiple survey waves. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys.

preliminary suggestive evidence of the school reform's likely impact on energy poverty by visualising its effects in Fig. 3. Here, we plot the local average levels of energy poverty, or the fraction of individuals classified as being energy poor against their month-year of birth, using January 1967 (represented by the dotted vertical line) as a reference point. The right side of each graph indicates exposure to the school reform, while the left side indicates non-exposure. We observe discontinuities in the energy poverty outcomes, which align with the timing of the exogenous increases in educational attainment, a trend consistent across all energy poverty measures. Specifically, we find that individuals exposed to the school reform tend to exhibit lower energy poverty scores or a reduced risk of energy poverty compared to their unexposed counterparts. We also provide preliminary evidence of a clear association between educational attainment and energy poverty in Fig. A6. We note a clear negative relationship between energy poverty and educational attainment, with energy poverty prevalence appearing much lower among those exposed to school reform. While this visualisation helps discern the differences in energy poverty scores or risk between the exposed and unexposed groups, it does not quantify, for example, the effects of a one-year increase in schooling on the risk of energy poverty. Therefore, we turn our attention to regression estimates generated using eq. (3), reporting the OLS, RF, and 2SLS estimates across various age bandwidths from 10 to 5 years on either side of the cut-off point to demonstrate the robustness of our findings.

Table 3 presents our main estimates of the impact of education on energy poverty, using MEPI status and MEPI score as primary measures for energy poverty. In analysing binary outcome variables, we employ linear probability models (LPM) in line with Angrist and Pischke (2009), owing to their clarity and practicality in estimating marginal effects. Table 3, Panel A, details the results from these models with MEPI status as the dependent variable, while Panel B focuses on MEPI score as the outcome variable. The uppermost section of each panel presents the OLS estimates, utilising years of schooling as the independent variable. The subsequent section presents estimates from the RF treatment effects of being 13 years or younger in 1980 or born after January 1967. The IV estimates, or fuzzy RD treatment effect estimates, are generated using the policy exposure dummy as an instrumental variable for schooling. The OLS estimates consistently reveal a negative, strong, and statistically significant correlation between years of schooling and energy poverty. The RF estimates in Panel A demonstrate a statistically significant negative impact of the school reform on MEPI status across all bandwidths except for the 5-year bandwidth. Specifically, exposure to the reform is associated with a reduction in energy poverty risk, ranging from approximately 12.00 pp. in broader bandwidths of 8 to 10 years to approximately 6.20 pp. in the narrower bandwidth of 6 years on either side of the cut-off, all statistically significant at the 1% and 5% levels, respectively. When considering the MEPI score, the RF estimates exhibit a similar pattern, albeit with slightly diminished robustness in a narrower bandwidth of 5 years. The IV estimates in panel A align with both OLS and RF estimates, indicating that a one-year increase in schooling corresponds to a significant decrease in the probability of energy poverty, ranging from 5.60 pp. in the 9-year bandwidth to 3.60 pp. in the

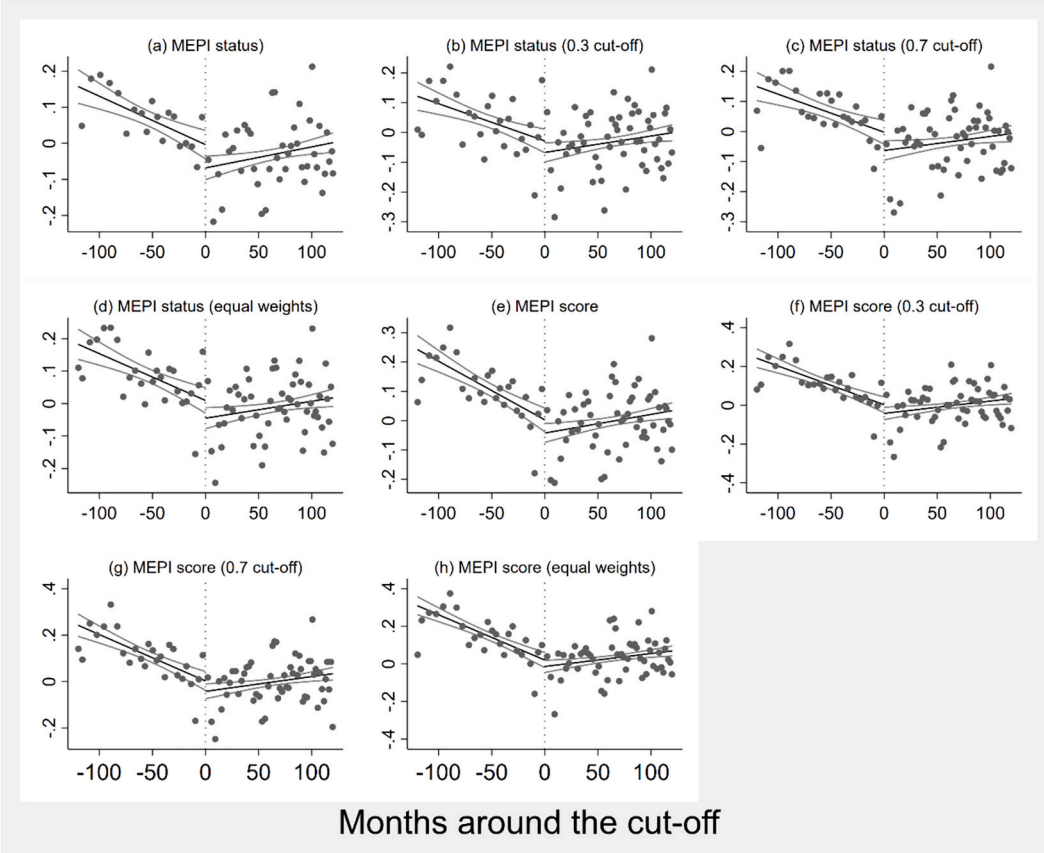

Fig. 3. The RD treatment effect of the 1980 school reform on educational attainment of men and women in Zimbabwe. The panels provide plots of the average educational attainment in monthly bins against the month-year of birth of being born in 1967. The point of discontinuity is centred on individuals born in 1967 – those who were 13 years old at the time of the school reform in 1980. The x-axis measures the number of months relative to this pivotal year when the respondent was 13 years old. Gray lines represent the 95% confidence intervals. The complete definitions of the variables of interest are provided in appendix. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys.

6-year bandwidth, statistically significant at the 1% level. These reductions in energy poverty risk, considering the average risk across respective bandwidths, translate to an approximate 8.56 to 5.53% decrease in the likelihood of experiencing energy poverty several years later in the life course. Additionally, we show that our primary estimates are also robust to alternative measures of energy poverty (Tables 5 and A6) as well as when we use the dummy variable for secondary education (Table A7).

Table 4 presents our analysis of the heterogeneous effects of education on energy poverty, highlighting potential differences by the gender of the household head and place of residence. Focusing on the estimates from the IV (fuzzy RD), our analysis reveals a differentiated impact of education on energy poverty in Zimbabwe based on head of household status and rural-urban samples. For individuals from male-headed households, a one-year increase in education mitigates energy poverty, indicated by a reduction of 5.6 to 4.9 pp. in the broader bandwidths of 10 to 8 years, respectively. This impact diminishes in the narrower bandwidths of 7 to 5 years, ranging from 3.0 to 2.1 pp This finding contrasts with the outcomes for individuals from female-headed households, where a one-year increase in schooling corresponds to a relatively lower decrease of 4.2 to 4.0 pp in the 10 to 8-year bandwidths. This effect persists with a negative and statistically significant impact on energy poverty in the narrower bandwidths of 5 to 7 years, with the largest effect observed in the 5-year bandwidth. In these narrower bandwidths, a one-year increase in education correlates with a reduction in energy poverty by 6.1 to 3.9 pp. Furthermore, the analysis delineates a notable contrast between urban and rural communities. In urban areas, the impact of education on energy poverty varies. While broader bandwidths (10 to 7 years) show a negative effect, narrower bandwidths (6 to 5 years) reveal a positive and statistically significant impact, with a one-year increase in schooling leading to a 6.4 pp increase in energy poverty in the 6-year bandwidth. Conversely, rural areas consistently exhibit a pronounced negative effect. Specifically, in rural areas, an additional year of schooling results in a 5.7 to 4.9 pp decrease in energy poverty within the 5 to 10-year bandwidths, significant at the 1% level (Table 4).

The results from Table 5 provide robustness checks for the impact of education on alternative measures of energy poverty in Zimbabwe, using different cut-off points for the MEPI status, which is a dummy variable reflecting energy poverty. These results are consistent with those found in Table 3, reinforcing the robustness of our main findings. For the MEPI status with a 0.3 cut-off, the IV estimates show that a one-year increase in schooling results in a statistically significant reduction in energy poverty, ranging from 3.4 to 1.5 pp. across broader bandwidths (10 to 6 years). However, in the 5-year bandwidth, the effect becomes positive and statistically significant at 4.6 pp. For the MEPI status with a 0.7 cut-off, the IV estimates indicate a consistently negative and statistically significant effect of education on energy poverty, ranging from 6.5 to 5.6 pp. across all bandwidths. For the MEPI status using equal weights (0.5 cut-off), the IV estimates also show a negative and statistically significant effect, with a one-year increase in schooling resulting in reductions ranging from 5.5 to 1.5 pp. across broader bandwidths (10 to 5 years). However, the coefficient in the 5-year bandwidth is not statistically significant. Given that our main estimates are based on several bandwidths, chosen using a parametric procedure which has been used in the previous related literature (;

Table 3
Effects of education on multidimensional energy poverty in Zimbabwe.

	Number of years on either side of the age cut-off											
	10		9		8		7		6		5	
Dependent variable: MEPI status												
OLS estimates												
Years of schooling	−0.076***	(0.004)	−0.077***	(0.004)	−0.077***	(0.004)	−0.075***	(0.003)	−0.075***	(0.002)	−0.075***	(0.001)
Observations	9787		8811		7721		6634		5612		4540	
Reduced form estimates												
Exposure to school reform	−0.121***	(0.012)	−0.115***	(0.021)	−0.115***	(0.031)	−0.070***	(0.018)	−0.062**	(0.022)	−0.032	(0.024)
Observations	9787		8811		7721		6634		5612		4540	
IV (fuzzy RD) estimates												
Years of schooling	−0.060***	(0.000)	−0.056***	(0.004)	−0.055***	(0.010)	−0.040***	(0.006)	−0.036***	(0.010)	−0.019	(0.012)
Observations	9787		8811		7721		6634		5612		4540	
Mean of the dependent variable	0.6519		0.6543		0.6530		0.6551		0.6510		0.6413	
First stage F-statistic [p-value]	108.372	[0.016]	77.455	[0.017]	109.373	[0.016]	92.910	[0.011]	124.337	[0.010]	31.395	[0.008]
A. Dependent variable: MEPI score												
OLS estimates												
Years of schooling	−0.088***	(0.003)	−0.088***	(0.003)	−0.089***	(0.003)	−0.087***	(0.002)	−0.087***	(0.001)	−0.087***	(0.001)
Observations	9787		8811		7721		6634		5612		4540	
Reduced form estimates												
Exposure to school reform	−0.123***	(0.008)	−0.115***	(0.018)	−0.114***	(0.029)	−0.086***	(0.020)	−0.090***	(0.017)	−0.030	(0.023)
Observations	9787		8811		7721		6634		5612		4540	
IV (fuzzy RD) estimates												
Years of schooling	−0.061***	(0.002)	−0.056***	(0.003)	−0.055***	(0.009)	−0.049***	(0.006)	−0.052***	(0.005)	−0.018	(0.011)
Observations	9787		8811		7721		6634		5612		4540	
First stage F-statistic [p-value]	108.372	[0.016]	77.455	[0.017]	109.373	[0.016]	92.910	[0.011]	124.337	[0.010]	31.395	[0.008]

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are shown in parentheses and clustered at the region and age cohort in 1980 levels to account for potential variation in the intensity of the school reform across the ten provinces or regions of the country. Individuals aged 14–15 in 1980 were partially exposed to the school reform, hence excluded from the main analysis sample and consistent with previous research using the same identification strategy. All regressions also include the following variables as additional controls: pre-policy and post-policy linear slopes, indicators for being female head of the household, affiliation to the apostolic church, employment in agriculture, married, access to piped water at the household, urban residence, household size fixed effects, survey year fixed effects, and region-age cohort in 1980 fixed effects. All regressions are weighted using sample probability weights. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys (DHS).

Makate and Nyamuranga, 2024), we provide further robustness checks to our estimates to the choice of the bandwidth using a data driven procedure proposed by Calonico et al. (2014). The optimal bandwidth, while varying for each specification, is somewhat narrower when compared to the others used in the parametric procedure described. However, these optimal bandwidths are very close to each other and encapsulated in the parametric procedure described. We report these results as supplementary materials in Table A13. The results are all consistent with our primary estimates. Overall, these robustness checks confirm that the negative impact of education on energy poverty observed in Table 3 remains valid under various alternative measures of energy poverty, affirming our findings' reliability and robustness.

6. Potential mechanisms linking education to energy poverty

The broader economic literature frequently hypothesises how education influences both market outcomes (such as employment and income) and non-market outcomes (such as health, social capital, and energy poverty, among others), which affect overall well-being and quality of life (Grossman, 2006). Higher educational attainment will likely improve labour market outcomes and household economic status, expand knowledge access, and modify societal norms and interpersonal dynamics. These mechanisms will likely enhance socioeconomic conditions and interactions with energy resources (Duflo, 2012). Grossman (2006) also discusses the broader channels of productive and allocative efficiency mechanisms, suggesting that education enhances individuals' efficiency in producing health and other non-market goods. Furthermore, education increases political and civic engagement, leading to calls for better energy regulations (Dee, 2004). Financial literacy gained through education allows for better household financial management and investment in energy-efficient equipment and infrastructure (Lusardi and Mitchell, 2014). Moreover, education fosters entrepreneurship by equipping individuals with the knowledge and skills to effectively access and utilise financial resources, enhancing energy access and efficiency (Van der Sluis et al., 2008).

6.1. Changes in labour market outcomes

Education significantly enhances labour market outcomes by increasing occupational opportunities and wage potential (Oreopoulos and Salvanes, 2011). Higher educational attainment equips individuals with advanced skills and knowledge, making them more competitive in the job market and enabling them, particularly women, to secure more stable and higher-paying jobs (; Whitmore Schanzenbach et al., 2016). This increased competitiveness reduces reliance on unstable employment forms such as casual or seasonal work, leading to increased

Table 4
Heterogeneous effects of the impact of education on energy poverty in Zimbabwe.

	Number of years on either side of the age cut-off											
	10		9		8		7		6		5	
Female headed household subsample												
OLS estimates												
Years of schooling	−0.045***	(0.007)	−0.046***	(0.007)	−0.047***	(0.006)	−0.044***	(0.006)	−0.046***	(0.005)	−0.051***	(0.006)
Observations	3540		3208		2839		2470		2117		1713	
Reduced form estimates												
Exposure to school reform	−0.075***	(0.015)	−0.092***	(0.008)	−0.074***	(0.006)	−0.071***	(0.002)	−0.103***	(0.004)	−0.088***	(0.022)
Observations	3540		3208		2839		2470		2117		1713	
IV (fuzzy RD) estimates												
Years of schooling	−0.042***	(0.005)	−0.050***	(0.001)	−0.040***	(0.002)	−0.039***	(0.003)	−0.052***	(0.003)	−0.061**	(0.022)
Observations	3540		3208		2839		2470		2117		1713	
Male headed household subsample												
OLS estimates												
Years of schooling	−0.050***	(0.003)	−0.050***	(0.004)	−0.051***	(0.004)	−0.049***	(0.006)	−0.047***	(0.007)	−0.046***	(0.005)
Observations	6156		5512		4827		4106		3464		2794	
Reduced form estimates												
Exposure to school reform	−0.115***	(0.031)	−0.105***	(0.023)	−0.110***	(0.018)	−0.063***	(0.017)	−0.015	(0.013)	−0.049***	(0.007)
Observations	6156		5512		4827		4106		3464		2794	
IV (fuzzy RD) estimates												
Years of schooling	−0.056**	(0.017)	−0.050***	(0.014)	−0.049***	(0.009)	−0.030***	(0.008)	−0.008	(0.006)	−0.021***	(0.004)
Observations	6156		5512		4827		4106		3464		2794	
Urban subsample												
OLS estimates												
Years of schooling	−0.053***	(0.001)	−0.053***	(0.001)	−0.053***	(0.003)	−0.049***	(0.004)	−0.045***	(0.007)	−0.048***	(0.006)
Observations	3306		2935		2592		2218		1864		1539	
Reduced form estimates												
Exposure to school reform	−0.095***	(0.008)	−0.078***	(0.010)	−0.049***	(0.013)	−0.008***	(0.001)	0.061***	(0.005)	0.035	(0.020)
Observations	3306		2935		2592		2218		1864		1539	
IV (fuzzy RD) estimates												
Years of schooling	−0.084***	(0.008)	−0.069***	(0.002)	−0.041**	(0.013)	−0.007***	(0.001)	0.064***	(0.000)	0.053	(0.036)
Observations	3306		2935		2592		2218		1864		1539	
Rural subsample												
OLS estimates												
Years of schooling	−0.045***	(0.002)	−0.045***	(0.001)	−0.046***	(0.001)	−0.045***	(0.001)	−0.045***	(0.000)	−0.046***	(0.002)
Observations	6390		5785		5074		4358		3717		2968	
Reduced form estimates												
Exposure to school reform	−0.119***	(0.033)	−0.129***	(0.030)	−0.134***	(0.028)	−0.121***	(0.023)	−0.129***	(0.016)	−0.167***	(0.009)
Observations	6390		5785		5074		4358		3717		2968	
IV (fuzzy RD) estimates												
Years of schooling	−0.049**	(0.016)	−0.050**	(0.016)	−0.052***	(0.015)	−0.048***	(0.012)	−0.051***	(0.008)	−0.057***	(0.006)
Observations	6390		5785		5074		4358		3717		2968	

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are shown in parentheses and clustered at the region and age cohort in 1980 levels to account for potential variation in the intensity of the school reform across the ten provinces or regions of the country. Individuals aged 14–15 in 1980 were partially exposed to the school reform, hence excluded from the main analysis sample and consistent with previous research using the same identification strategy. All regressions also include the following variables as additional controls: pre-policy and post-policy linear slopes, indicators for being female head of the household, affiliation to the apostolic church, employment in agriculture, married, access to piped water at the household, urban residence, household size fixed effects, survey year fixed effects, and region-age cohort in 1980 fixed effects. All regressions are weighted using sample probability weights. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys (DHS).

financial stability and improved outcomes (OECD, 2022; Whitmore Schanzenbach et al., 2016) including access to modern energy services. Increased employability in formal sectors contributes to better job security and income stability, essential for consistent energy access and reducing energy poverty (OECD, 2022). Stable income streams allow households to invest in energy-efficient technologies and reliable energy sources, further mitigating energy poverty (de Ia Rue du Can and Price, 2008). Additionally, higher education levels within a population can lead to a more skilled workforce, fostering economic growth and stability, which enhances energy infrastructure and availability at a national level (Duflo, 2012). Education also promotes innovation and adaptability among workers, making them more resilient to economic shocks and capable of transitioning to new energy-efficient technologies and practices (Becker, 1975). Thus, education improves labour market outcomes by increasing employability, job stability, and wage potential, which are crucial for securing reliable energy access and contributing to broader economic stability and development, thereby reducing energy poverty.

We test this potential mechanism and provide results in Table 6, generated using Eq. (3). In Eq. (3), we replace the outcome variable (energy poverty) with each of the mechanisms considered in this paper. The OLS estimates indicate that increased schooling negatively correlates with the probability of self-employment, agricultural, seasonal, casual, and domestic work while positively associated with current, consistent, or year-round and professional/managerial employment. The RF estimates show that school reform exposure increases the probability of self-employment by 1.9 pp. ($p < 0.01$), current employment by 2.4 pp. ($p < 0.01$), and year-round employment by 5.5 pp. ($p < 0.01$). Also, school reform exposure reduces the prospect of seasonal work by 1.1 pp. ($p < 0.05$), casual work by 1.7 pp. ($p < 0.01$),

Table 5
Robustness checks: effects of education on alternative measures of energy poverty.

	Number of years on either side of the age cut-off											
	10		9		8		7		6		5	
Dependent variable: MEPI status (0.3 cut-off)												
OLS estimates												
Years of schooling	−0.075***	(0.004)	−0.074***	(0.004)	−0.074***	(0.004)	−0.072***	(0.002)	−0.071***	(0.000)	−0.071***	(0.001)
Observations	9787		8811		7721		6634		5612		4540	
Reduced form estimates												
Exposure to school reform	−0.069***	(0.018)	−0.053*	(0.024)	−0.053	(0.033)	−0.025	(0.017)	−0.025	(0.014)	0.076***	(0.012)
Observations	9787		8811		7721		6634		5612		4540	
IV (fuzzy RD) estimates												
Years of schooling	−0.034***	(0.006)	−0.026**	(0.009)	−0.026	(0.014)	−0.014	(0.008)	−0.015*	(0.007)	0.046**	(0.015)
Observations	9787		8811		7721		6634		5612		4540	
First stage F-statistic [p-value]	108.372	[0.016]	77.455	[0.017]	109.373	[0.016]	92.910	[0.011]	124.337	[0.010]	31.395	[0.008]
Dependent variable: MEPI status (0.7 cut-off)												
OLS estimates												
Years of schooling	−0.076***	(0.001)	−0.077***	(0.002)	−0.076***	(0.001)	−0.074***	(0.001)	−0.075***	(0.000)	−0.074***	(0.001)
Observations	9787		8811		7721		6634		5612		4540	
Reduced form estimates												
Exposure to school reform	−0.131***	(0.012)	−0.125***	(0.021)	−0.136***	(0.028)	−0.099***	(0.024)	−0.119***	(0.027)	−0.093***	(0.026)
Observations	9787		8811		7721		6634		5612		4540	
IV (fuzzy RD) estimates												
Years of schooling	−0.065***	(0.000)	−0.061***	(0.003)	−0.066***	(0.007)	−0.056***	(0.008)	−0.069***	(0.010)	−0.056***	(0.006)
Observations	9787		8811		7721		6634		5612		4540	
First stage F-statistic [p-value]	108.372	[0.016]	77.455	[0.017]	109.373	[0.016]	92.910	[0.011]	124.337	[0.010]	31.395	[0.008]
Dependent variable: MEPI status (using equal weights)												
OLS estimates												
Years of schooling	−0.081***	(0.003)	−0.081***	(0.004)	−0.080***	(0.003)	−0.079***	(0.002)	−0.078***	(0.001)	−0.080***	(0.000)
Observations	9787		8811		7721		6634		5612		4540	
Reduced form estimates												
Exposure to school reform	−0.111***	(0.021)	−0.097***	(0.024)	−0.106**	(0.033)	−0.088***	(0.024)	−0.096***	(0.026)	−0.025	(0.025)
Observations	9787		8811		7721		6634		5612		4540	
IV (fuzzy RD) estimates												
Years of schooling	−0.055***	(0.005)	−0.047***	(0.007)	−0.051***	(0.011)	−0.050***	(0.008)	−0.056***	(0.011)	−0.015	(0.013)
Observations	9787		8811		7721		6634		5612		4540	
First stage F-statistic [p-value]	108.372	[0.016]	77.455	[0.017]	109.373	[0.016]	92.910	[0.011]	124.337	[0.010]	31.395	[0.008]

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are shown in parentheses and clustered at the region and age cohort in 1980 levels to account for potential variation in the intensity of the school reform across the ten provinces or regions of the country. Individuals aged 14–15 in 1980 were partially exposed to the school reform, hence excluded from the main analysis sample and consistent with previous research using the same identification strategy. All regressions also include the following variables as additional controls: pre-policy and post-policy linear slopes, indicators for being female head of the household, affiliation to the apostolic church, employment in agriculture, married, access to piped water at the household, urban residence, household size fixed effects, survey year fixed effects, and region-age cohort in 1980 fixed effects. All regressions are weighted using sample probability weights. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys (DHS).

agricultural work by 1.6 pp. ($p < 0.01$), skilled manual work by 1.3 pp. ($p < 0.01$), and domestic work by 0.7 pp. ($p < 0.01$). The IV estimates show a similar pattern to the RF estimates, indicating that a one-year increase in schooling increases the chances of self-employment by 1.1 pp. ($p < 0.01$), current employment by 1.3 pp. ($p < 0.01$), and full-year employment by 3.1 pp. ($p < 0.01$). An additional year of schooling reduces the chances of being a seasonal worker by 0.6 pp. ($p < 0.01$), causal worker by 1.0 pp. ($p < 0.01$), agricultural worker by 0.9 pp. ($p < 0.01$), skilled manual worker by 0.7 pp. ($p < 0.01$), and domestic work by 0.4 pp. ($p < 0.01$). Our results remain robust even when we consider the dummy variable for secondary school completion as the primary independent variable in place of years of schooling (see Table A9). These findings highlight education's pivotal influence in improving labour market outcomes, leading individuals to more secure and financially rewarding employment. This development is essential for economic progress and helps reduce energy poverty. Education's influence on reshaping employment opportunities is a crucial mechanism in our study, offering suggestive evidence of its vital role in fostering sustainable economic progress and alleviating energy poverty.

6.2. Changes in household economic and infrastructure indicators

Changes in household economic and infrastructure indicators play a crucial role in alleviating energy poverty. As economic theory suggests, educational attainment elevates household economic status and infrastructure access, critical enabling factors in reducing energy poverty. By

Table 6
Effects of education on labor market outcomes.

	Self employed	Currently employed	Employed all year	Seasonal worker	Casual worker	Agricultural worker	Professional/ managerial work	Skilled manual work	Domestic worker
OLS estimates									
Years of schooling	−0.001*	0.020***	0.024***	−0.006***	−0.001***	−0.009***	0.035***	−0.002***	−0.003***
	(0.001)	(0.000)	(0.001)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)
Observations	16,396	16,396	16,396	16,396	16,396	16,396	16,396	16,396	16,396
Reduced form estimates									
Exposure to school reform	0.019***	0.024***	0.055***	−0.011**	−0.017***	−0.016***	0.028	−0.013***	−0.007***
	(0.002)	(0.003)	(0.004)	(0.004)	(0.000)	(0.001)	(0.015)	(0.000)	(0.002)
Observations	16,396	16,396	16,396	16,396	16,396	16,396	16,396	16,396	16,396
IV (fuzzy RD) estimates									
Years of schooling	0.011***	0.013***	0.031***	−0.006**	−0.010***	−0.009***	0.016	−0.007***	−0.004***
	(0.001)	(0.001)	(0.003)	(0.002)	(0.000)	(0.000)	(0.009)	(0.000)	(0.001)
Observations	16,396	16,396	16,396	16,396	16,396	16,396	16,396	16,396	16,396
First stage F-statistic	746.9	746.9	746.9	746.9	746.9	746.9	746.9	746.9	746.9
P-value	0.0130	0.0130	0.0130	0.0130	0.0130	0.0130	0.0130	0.0130	0.0130

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are shown in parentheses and clustered at the region and age cohort in 1980 levels to account for potential variation in the intensity of the school reform across the ten provinces or regions of the country. Individuals aged 14–15 in 1980 were partially exposed to the school reform, hence excluded from the main analysis sample and consistent with previous research using the same identification strategy. All regressions also include the following variables as additional controls: pre-policy and post-policy linear slopes, indicators for being female, household size and household size squared, urban residence, survey year fixed effects, and region fixed effects. All regressions are weighted using sample probability weights that have been adjusted to account for pooling across multiple waves of the Zimbabwe DHS. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys (DHS).

improving human capital, education boosts the likelihood of asset accumulation and improves household infrastructure, including advanced sanitation and water sources (Behrman et al., 2012; Glewwe, 2002; Poterba et al., 2013). These improvements, in turn, enhance living standards and access to essential services, facilitating better energy resource management (Campbell et al., 2003; Khandker et al., 2012). Improved economic status allows households to invest in cleaner, more efficient energy sources, reducing reliance on traditional, less efficient energy sources, and mitigating energy poverty (Dinkelman, 2011). Higher educational attainment boosts household incomes, enabling families to afford better housing and infrastructure, which facilitates investments in modern energy systems such as solar panels, efficient cooking stoves, and reliable electricity connections, significantly reducing energy poverty (Bernard, 2012; Wolfram et al., 2012). Education equips individuals with the knowledge and skills needed to access and utilise financial services effectively, enhancing their ability to secure loans and invest in home improvements and energy infrastructure (Dupas and Robinson, 2013). This economic empowerment helps households transition from traditional biomass fuels to modern energy sources, which are cleaner, more efficient, and less harmful to health (Pachauri et al., 2012).

Our analysis investigates this potential mechanism, presenting results in Table 7 that explore the impact of education on household asset wealth, access to piped water, and improved sanitation. We report the OLS, RF and IV estimates for each outcome variable generated using Eq. (3), with each mechanism as the dependent variable. The OLS estimates indicate a positive and statistically significant association between education and the probability of having high asset wealth (i.e., belonging to the highest household asset quintiles 4 and 5) and improved toilet facilities but negatively correlates with access to piped water and low asset wealth (i.e., belonging to the lowest household asset quintiles 1 and 2). The RF estimates further reveal that exposure to school reform significantly increases the probability of possessing high asset wealth by 3.6 pp. ($p < 0.01$), enhanced toilet facilities by 3.8 pp. ($p < 0.01$), and piped water by 6.3 pp. ($p < 0.01$). Additionally, exposure to school reform reduces the chances of low-asset wealth by 3.1 pp. ($p < 0.01$). The IV estimates show a consistent pattern indicating that a one-year increase in schooling increases the likelihood of high asset wealth by 2.1 pp. ($p < 0.01$), improved toilet by 2.2 pp. ($p < 0.01$), and piped water by 3.6 pp. ($p < 0.01$). An additional year in school lowers the risk of having a low asset wealth by an estimated 3.1 pp. ($p < 0.01$). Our results remain

Table 7
Effect of education on household economic and infrastructure indicators.

	High asset wealth	Low asset wealth	Improved toilet	Piped water
OLS estimates				
Years of schooling	0.028***	−0.027***	0.026***	−0.006***
	(0.001)	(0.000)	(0.000)	(0.001)
Observations	16,396	16,396	16,396	16,396
Reduced form estimates				
Exposure to school reform	0.036***	−0.054***	0.038***	0.063***
	(0.003)	(0.009)	(0.002)	(0.017)
Observations	16,396	16,396	16,396	16,396
IV (fuzzy RD) estimates				
Years of schooling	0.021***	−0.031***	0.022***	0.036***
	(0.001)	(0.004)	(0.000)	(0.008)
Observations	16,396	16,396	16,396	16,396
First stage F-statistic	573.4	573.4	573.4	573.4
P-value	0.0126	0.0126	0.0126	0.0126

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Estimates are generated using the preferred bandwidth comprising of individuals who were aged 5–24 years in 1980 but excludes those who were partially exposed to the reform. Standard errors are shown in parentheses and clustered at the region and age cohort in 1980 levels to account for potential variation in the intensity of the school reform across the ten provinces or regions of the country. In all the regressions, we include additional controls for pre-policy and post-policy linear trends, household size and household size squared, survey year fixed effects, and region fixed effects. All regressions are weighted using sample probability weights that account for pooling across multiple surveys. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys (DHS).

robust even when we consider the dummy variable for secondary school completion as the primary independent variable in place of years of schooling (see Table A10). These findings reveal that education not only equips individuals with knowledge but also positively influences economic and infrastructural development, contributing to improved living standards and sustainable development.

6.3. Improved access to information

Education transcends individual learning, crucially contributing to broader community access to information and shaping behavioural norms. Education improves access to information (Makate and Nyamuranga, 2023), which is crucial for adopting energy-efficient technologies and practices. Educated individuals are more likely to be aware of and adopt energy-saving options, contributing to more sustainable energy consumption patterns (Mills and Schleich, 2012). Enhanced access to information enables individuals to make informed decisions about energy use, understand the benefits of energy efficiency, and access financial mechanisms to invest in energy-saving technologies (Gillingham et al., 2009). This broader dissemination of knowledge within the community fosters collective improvements in energy practices, thereby reducing energy poverty (Rao and Pachauri, 2017). Additionally, educated individuals are better equipped to utilise governmental and non-governmental programs to improve energy access and efficiency, mitigating energy poverty (Peters and Sievert, 2016). By enhancing the flow of information, education empowers people and communities to adopt sustainable energy solutions, contributing to long-term environmental conservation and economic stability.

Table 8 presents the results that explore this potential channel. The OLS estimates reveal a positive and significant correlation between years of schooling and access to information through frequently reading newspapers or magazines, listening to the radio, watching television, and living close to a neighbor with better access to information through radio or television. The RF estimates also indicate a positive and significant correlation between school reform and access to information, except for the probability of watching television. Additionally, the IV estimates show that an extra year of schooling increases the probability of reading newspapers frequently by 8.4 pp. ($p < 0.01$), listening to the radio frequently by 3.0 pp. ($p < 0.01$), living close to a neighbor with access to a television and radio by 2.1 pp. ($p < 0.01$) and 0.6 pp. ($p < 0.01$), respectively. This evidence suggests that education not only enhances individual access to information but also positively influences the informational environment within the community. The association between education and neighbours having access to a radio or television implies that educated individuals may reside in better-connected communities where information flows more freely. This connectivity could further facilitate knowledge sharing and collective actions towards energy efficiency and sustainable development. Our results remain robust even when we consider the dummy variable for secondary school completion as the primary independent variable in place of years of schooling (see Table A11). These findings underscore the pivotal role of education in promoting energy efficiency and sustainable development. Education can drive positive change at the community level by providing individuals with the knowledge and resources necessary to save energy, lifting people out of various forms of poverty including energy poverty.

Table 8
Effects of education on access to information.

	Reads newspapers	Listens to the radio	Watches television	Neighbor has access to TV	Neighbor has access to the radio
OLS estimates					
Years of schooling	0.064***	0.031***	0.045***	0.022***	0.010***
	(0.001)	(0.000)	(0.000)	(0.001)	(0.000)
Observations	16,393	16,391	16,386	16,382	16,385
Reduced form estimates					
Exposure to school reform	0.162***	0.059***	0.038	0.042***	0.011***
	(0.012)	(0.014)	(0.023)	(0.003)	(0.001)
Observations	16,393	16,391	16,386	16,382	16,385
IV (fuzzy RD) estimates					
Years of schooling	0.084***	0.030***	0.020	0.021***	0.006***
	(0.001)	(0.005)	(0.011)	(0.000)	(0.001)
Observations	16,393	16,391	16,386	16,382	16,385
First stage F-statistic	314.0	313.4	322.8	309.1	309.0
P-value	0.0142	0.0142	0.0142	0.0143	0.0143

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Estimates are generated using the preferred bandwidth comprising of individuals who were aged 5–24 years in 1980 but excludes those who were partially exposed to the reform. Standard errors are shown in parentheses and clustered at the region and age cohort in 1980 levels to account for potential variation in the intensity of the school reform across the ten provinces or regions of the country. In all the regressions, we include additional controls for pre-policy and post-policy linear trends, household size and household size squared, survey year fixed effects, and region fixed effects. All regressions are weighted using sample probability weights that account for pooling across multiple surveys. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys (DHS).

6.4. Changes in social norms and interpersonal dynamics

Education significantly impacts social norms and interpersonal dynamics - defined as the interactions and relationships among individuals within a household or community - especially concerning gender roles and intimate partner violence, which has direct implications for energy poverty. Education empowers women by enhancing cognitive abilities and critical thinking, fostering equitable household decision-making and resource management (Duflo, 2012; Sen, 1999). This empowerment reduces tolerance for violence against women and promotes gender equality within households, improving energy management and reducing energy poverty (Heath and Mushfiq Mobarak, 2015; Makate and Nyamuranga, 2024). Educated women are more likely to participate in crucial household decisions related to energy consumption. Their involvement results in more informed and sustainable energy practices, enhancing overall household welfare and sustainability (Kabeer, 2005). Economic theory posits that education reshapes beliefs and practices, fostering a more harmonious household environment and better energy resource management (Becker, 2009; Schultz, 1961). Furthermore, education fosters greater social capital, facilitating collective action in communities to improve energy infrastructure and access (Huang et al., 2009). These dynamics demonstrate how education empowers individuals and catalyses broader social and economic benefits that contribute to long-term reductions in energy poverty.

We investigate this potential mechanism by examining whether increased education has an impact on tolerance for violence against women and report the results in Table 9. The OLS estimates reveal a negative and statistically significant association between years of schooling and tolerance for violence. The RF estimates also indicate a negative effect of school reform and outcomes indicative of tolerance for violence against women. The IV estimates suggest that a one-year of schooling reduces overall tolerance for violence by 0.057. ($p < 0.01$), the probability of accepting that a husband is justified in beating a wife if she goes out without telling him (2.4 pp.; $p < 0.01$), neglects the children (2.2 pp.; $p < 0.01$), argues with him (1.7 pp.; $p < 0.01$), refuses sex with him (2.6 pp.; $p < 0.01$), and burns food (1.5 pp.; $p < 0.01$). Our results remain robust even when we consider the dummy variable for secondary school completion as the primary independent variable in

Table 9
Effects of education on tolerance for violent beliefs against women.

		Respondent believes partner violence is justifiable in the following circumstances:				
	Tolerance for violence against women	Goes out without telling him	Neglects the children	Argues with him	Refuses sex with him	Burns the food
OLS estimates						
Years of schooling	−0.035***	−0.016***	−0.014***	−0.015***	−0.014***	−0.007***
	(0.002)	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)
Observations	15,334	15,179	15,241	15,190	15,161	15,290
Reduced form estimates						
Exposure to school reform	−0.097***	−0.041***	−0.037***	−0.029**	−0.044***	−0.026***
	(0.003)	(0.002)	(0.006)	(0.011)	(0.004)	(0.007)
Observations	15,334	15,179	15,241	15,190	15,161	15,290
IV (fuzzy RD) estimates						
Years of schooling	−0.057***	−0.024***	−0.022***	−0.017**	−0.026***	−0.015**
	(0.001)	(0.000)	(0.004)	(0.005)	(0.001)	(0.005)
Observations	15,334	15,179	15,241	15,190	15,161	15,290
First stage F-statistic	257.8	306.9	253.1	259.1	583.5	435.3
P-value	0.0123	0.0121	0.0125	0.0121	0.0126	0.0125

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Estimates are generated using the preferred bandwidth comprising of individuals who were aged 5–24 years in 1980 but excludes those who were partially exposed to the reform. Standard errors are shown in parentheses and clustered at the region and age cohort in 1980 levels to account for potential variation in the intensity of the school reform across the ten provinces or regions of the country. In all the regressions, we include additional controls for pre-policy and post-policy linear trends, household size, survey year fixed effects, and region fixed effects. All regressions are weighted using sample probability weights that account for pooling across multiple surveys. Source: Data are from the 1999, 2005/2006, 2010/2011, and 2015 Zimbabwe demographic and health surveys (DHS).

place of years of schooling (see Table A12). Collectively, these findings underscore the transformative role of education in reshaping societal norms and promoting gender equity. This improvement in household dynamics, driven by increased education, enhances decision-making and resource management within households. Educated individuals are likely to make more informed and equitable decisions regarding the use of resources, including energy. These positive changes in household behaviour and attitudes can lead to more efficient energy use, reducing energy poverty. Education thus plays a critical role not only in promoting gender equity but also in fostering sustainable development and improving living standards through better resource management.

7. Discussion

This study explores the effect of education on energy poverty, leveraging the unique context provided by the 1980 school reform in Zimbabwe. This reform provided an exogenous increase in education levels, serving as an ideal natural experiment to causally examine how exposure to school reform influences the risk of energy poverty among female- and male-headed families in Zimbabwe. The evidence from the RD framework and using multiple waves of the Zimbabwe DHS collected in 1999, 2005/2006, 2010/2011, and 2015 suggests that increased education reduces energy poverty. Our findings align with existing literature that highlights the multifaceted role of education in poverty alleviation and economic development (Dagher et al., 2023; Hassan et al., 2022; Sun et al., 2023; Zhang et al., 2021).

We identified four crucial mechanisms through which education influences energy poverty: changes in labour market outcomes, alterations in household economic and infrastructure indicators, improvements in access to information, and shifts in social norms and interpersonal dynamics.

The first mechanism we identify pertains to education's role in improving labour market outcomes, leading to more stable and better-paid employment opportunities. However, the impact of education on energy poverty in Zimbabwe is complex and influenced by gender-specific employment patterns. This dynamic is crucial in understanding the differential impact of education on energy poverty. While our findings align with the broader economic literature that underscores the positive influence of education on employment opportunities (Duflo, 2012; Jia and Li, 2021; Wang et al., 2021), they reveal a significant gender disparity. A striking aspect of our findings is the substantial proportion of women who, despite higher educational gains, remain unemployed or are employed and predominantly occupy traditionally lower-wage roles such as sales, clerical, and domestic services. In contrast, men are more frequently employed in higher-paying professional, technical, managerial, and skilled manual jobs. We present the results of this additional analysis as supplementary material in Table A8. This gender-based occupational segregation helps explain why education's role in alleviating energy poverty is less pronounced in female-headed households than in male-headed households. Further exacerbating this situation are the prevailing societal norms in Zimbabwe, as evidenced by a higher societal tolerance for violence against women. These norms, deeply entrenched in the patriarchal fabric of society, not only reinforce gender disparities in the labour market but also impede the full realisation of education's potential in economic empowerment of women. Despite similar educational achievements, the disparities in asset ownership between male-headed and female-headed households point to a broader systemic issue. Female-headed families' limited asset wealth curtails their capacity to invest in energy management, highlighting the need for policies that simultaneously address educational and gender inequalities. This observation aligns with findings by Sun et al. (2023), who emphasise the role of education in fostering innovation and financial development, key to reducing energy poverty.

The second mechanism we identified is the impact of education on household economic status and infrastructure. Our analysis revealed that increased educational attainment brought about by exposure to school reform in Zimbabwe positively correlates with higher asset wealth and improved living standards, as evidenced by ownership of improved toilet facilities and access to piped water, while negatively associated with low asset wealth. This connection between education and economic upliftment is highlighted in the broader economic literature (Behrman et al., 2012; Campbell et al., 2003; Glewwe, 2002; Khandker et al., 2012; Poterba et al., 2013). Hassan et al. (2022) discuss the interplay between globalization, education, and income inequalities in addressing energy poverty, supporting our findings that higher education levels correlate with better economic outcomes and lower energy

poverty. However, it must be contextualised within Zimbabwe's unique policy environment. Economic challenges in the late 1990s and early 2000s, including hyperinflation and political instability, impacted educational quality and access, particularly in rural areas (Government of Zimbabwe, 2020; Kanyongo, 2005). The government's early 2000s land reform program, intended to address historical inequalities, inadvertently influenced agricultural productivity and household income (Jombo et al., 2017). Recent policy initiatives, such as the Transitional Stabilisation Program (TSP) and the National Development Strategy 1 (NDS1), aim to revitalise the economy and infrastructure (Government of Zimbabwe, 2020). However, the effectiveness of education in mitigating energy poverty is contingent upon effective policy implementation, adequate funding, and political stability. Thus, while education remains a significant factor in reducing energy poverty, its impact is mediated by these broader socioeconomic and political conditions. Our findings underscore the need for comprehensive policy strategies that enhance educational outcomes and address the country's underlying economic and infrastructural challenges. This supports the work of Dagher et al. (2023), who found that energy poverty is inversely related to the educational level of the household head, with higher education correlating with lower incidence of energy poverty due to higher income levels.

Our analysis indicates that increased education significantly enhances access to information in Zimbabwe, a critical factor in addressing energy poverty. This finding aligns with the broader economic literature, which consistently demonstrates the positive impact of education on information access (Makate and Nyamuranga, 2023; Mills and Schleich, 2012). Zhang et al. (2021) highlight the importance of educational attainment in enhancing children's well-being and academic performance, which indirectly affects energy poverty, reinforcing the critical role of information access in facilitating social change (Stiglitz, 2000). The government of Zimbabwe's focus on information and communications technologies (ICTs) in the National ICT Policy 2005 underscores the importance of ICT in sectors including education (Government of Zimbabwe, 2016; The World Bank, 2007). However, challenges such as inadequate ICT infrastructure and skills gaps, particularly in rural areas, are barriers to realising the full potential of this mechanism. The policy's objectives to foster growth in telecommunications and ICT use across sectors and to bridge the digital divide are crucial (Government of Zimbabwe, 2016). The emphasis on human resource skills, capacity building, and research is critical to enhancing information access through education (Government of Zimbabwe, 2016). However, education's role in enhancing access to energy information in Zimbabwe varies, influenced by considerable regional disparities in educational quality and technological infrastructure. Particularly in regions like Matabeleland, where there is a shortage of infrastructure, inadequate classrooms, and teacher shortages, the educational advantage in accessing energy-related information may be less pronounced (Pulitzer Center, 2023). Furthermore, issues such as poor network coverage and internet connectivity hinder the effectiveness of online learning strategies, which are critical in today's education landscape. Therefore, while education remains crucial in combating energy poverty, its efficacy in Zimbabwe depends on addressing these regional challenges and aligning policy efforts with the population's diverse needs.

Our examination of the interplay between education and energy poverty reveals another important mechanism. Education fundamentally alters social norms, significantly reducing tolerance for violence against women and reshaping gender dynamics. According to Sen (1999), education can improve cognitive skills and critical thinking, leading to fairer beliefs and practices within households. This observation is particularly significant in Zimbabwe, where there is a higher acceptance of violence against women (Cools and Kotsadam, 2017; Makate and Nyamuranga, 2024), making it harder for education to help women escape energy poverty compared to men. Duflo (2012) adds to this argument by pointing out that education empowers women, which changes household dynamics that are important for decision-making and resource management, ultimately helping to reduce energy poverty. In this context, it is important for Zimbabwe's educational policies to directly combat gender-based violence and enhance gender equality in both educational and economic sectors. These policies should provide education that not only raises awareness about gender equality but also equips women with the tools and opportunities for equal participation in economic decisions, including energy resource management. Such an approach is not only grounded in economic theories but is also tailored to meet the specific socio-economic needs of the country, thereby strengthening the impact of education on reducing energy poverty especially among women. This mechanism is further supported by Hassan et al. (2022), who discuss the interplay between globalization, education, and income inequalities in addressing energy poverty.

7.1. Strengths and limitations

Our study contributes significantly to understanding the long-term impact of education on energy poverty in Zimbabwe, offering robust empirical evidence through a natural experiment and an RD design. This approach ensures strong causal inference, particularly regarding the long-term effects of educational reforms. However, the study is subject to several limitations. First, the reliance on self-reported data from the Zimbabwe DHS may introduce reporting biases, especially concerning sensitive issues such as domestic violence. Second, while the RD design provides robust causal inference, it is limited to individuals near the cutoff age, potentially affecting the generalisability of our findings to the broader population. Third, our focus on secondary education may overlook the long-term impacts of primary and tertiary education on energy poverty outcomes. Additionally, Zimbabwe's unique socioeconomic and political conditions during the study period, such as economic instability and land reforms, may confound our results despite our efforts to control for regional and temporal variations. These limitations underscore the need for further research employing diverse contexts and considering a broader spectrum of educational levels to validate and expand our findings. Moreso, exploring longitudinal data across different regions and socioeconomic conditions will provide deeper insights into the long-term impacts of educational interventions on energy poverty outcomes. Such research will be instrumental in informing policy decisions to leverage educational improvements to alleviate energy poverty effectively.

8. Conclusions and policy suggestions

In this study, we explored the relationship between education and energy poverty, leveraging the exogenous variability introduced by the 1980 Zimbabwe school reform to identify the causal effect of education on energy poverty. Drawing on data from the Zimbabwe DHS spanning 1999 to 2015 and employing an RD design, we found a notable increase in years of schooling post-reform, about 2.08 years on average, with more pronounced effects among women (2.34 years) and rural residents (2.53 years). Our findings indicate that increased schooling reduces the likelihood of energy poverty by an estimated 5.61 to 8.56%. However, our findings reveal a notable gender disparity in the impact of increased education. Although women in general showed higher educational gains than men, male heads of households achieved higehr educational gains compared to female heads of households. Consequently, male-headed households experienced a greater reduction in energy poverty. This disparity is possibly attributable to observed gender inequalities in the labour market, where women are often unemployed or in lower-paying roles, and societal norms that disadvantage women, especially in the context of domestic violence. The complex interplay of these factors likely limits the effectiveness of education in reducing energy poverty among female-headed households in Zimbabwe. This evidence highlights the need for policies that address educational and broader societal

inequities.

Given the demonstrated impact of education on reducing energy poverty in Zimbabwe, through mechanisms such as information access, infrastructural improvements, and changes in societal norms, several policy implications emerge.

First, policy efforts should prioritise educational equity, aligning with initiatives like 'Education Plus', which empowers adolescent girls and young women (UNAIDS, 2023). These efforts should address gender disparities in education and the labour market, as our findings indicate a diminished effect of education on energy poverty among female-headed families. Tailoring educational policies to combat gender-based occupational segregation and economic challenges is crucial. Specific measures in Zimbabwe could include scholarships for girls in Science, Technology, Engineering, and Mathematics (STEM) fields, vocational training programs focused on high-demand sectors, and initiatives encouraging female entrepreneurship. Addressing societal norms that discourage women from pursuing certain careers can be achieved through awareness campaigns and mentorship programs highlighting successful female role models in various fields. Additionally, implementing or providing ongoing support to microfinance programs and entrepreneurship training for women can enhance their economic independence and opportunities. Such efforts are vital for mitigating energy poverty among this demographic, given our findings that education has a reduced impact on energy poverty among female-headed households.

Secondly, policymakers should enhance information dissemination infrastructure, particularly in rural areas with more pronounced disparities. Enhancing information dissemination requires boosting digital connectivity, addressing gaps in Information and Communication Technology (ICT) infrastructure, and increasing teacher capacity. Targeted initiatives include deploying mobile technology solutions and satellite internet services to overcome poor network coverage, alongside training programs for educators to utilise digital platforms for energy education effectively. These interventions could be tailored to the specific challenges of regions like Matabeleland North, Matabeleland South, and Manicaland, where infrastructure deficits are more acute (Pushak and Briceno-Garmendia, 2011). Partnerships with local telecommunication companies such as Econet Wireless and NetOne to provide affordable internet access, as well as the development of localised digital content in multiple languages, would be beneficial. These efforts would improve access to essential information, enhance educational outcomes, and support sustainable energy practices across Zimbabwe.

Lastly, altering social norms and interpersonal dynamics through community-based programs is essential for promoting energy-efficient practices and sustainable energy usage. These programs should focus on gender equality and empowerment, encouraging women's participation in household energy management and broader community energy initiatives. For Zimbabwe, this could involve collaboration with traditional leaders, local non-governmental organisations (NGOs), and government agencies to create culturally sensitive programs that promote gender equality and energy efficiency. Specific initiatives could include community energy forums where women can share knowledge and experiences, as well as training workshops on modern energy solutions such as solar cookers and energy-saving stoves. Partnering with local chiefs and village headmen to endorse and support these initiatives can enhance their acceptance and effectiveness, leveraging their influential roles in community governance across all provinces.

CRediT authorship contribution statement

Marshall Makate: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Appendix A. Supplementary data

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